Gaspar Humberto Valdez Avila

Bachelor of Science in Virtual Environments and Digital Business | Mérida, Yucatán, México | (+52) 999 246 8034 |
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Video Game and Immersive Environments Developer (Unity, Godot Engine) with over 7 years of experience in the creation of video games and immersive environments (VR, AR, and Metaverses). Specialized in real-time multiplayer systems, performance optimization, and interactive interface design for web and mobile platforms. Focused on rapid prototyping and continuous improvement through data analytics and Agile methodologies.

PROFESSIONAL EXPERIENCE

TrainBeyond LLC **Remote (Based in Mérida, Yucatán, México)**
Virtual Environments Developer / VR Environments *May 2026 – September 2026*

- Developed and integrated haptic and physics interaction systems in Virtual Reality (grabbable modules and tool manipulation), optimizing grabbing precision and user response in immersive environments.
- Programmed industrial operational simulation mechanics in C# / Unity, including the operation and optical scanning of mobile terminals (handhelds) for barcode/label reading, as well as cargo transportation using hydraulic pallet jacks.
- Designed autonomous heavy machinery logic, implementing automatic operation sequences and traffic/movement routines for forklifts within the virtual warehouse.
- Created a dynamic emergency procedural system with stochastic fire outbreak generation, integrating critical response mechanics, finite consumable logic for fire extinguishers, and physics-based mitigation conditions.
- Structured the state machine for risk detection and assessment, programming procedural failure conditions (failure scenarios / game overs) and industrial safety formative feedback criteria.
- Designed and implemented an immersive 3D spatial audio architecture, synchronizing sound feedback with mechanical and environmental alarm events to maximize situational realism.

UNIIDE / Universidad Tecnológica Metropolitana **Mérida Yucatán, México**
Immersive Environments Developer (VARTICA Metaverse) *January 2025 – April 2026*

- Developed a multiplatform web immersive environment using Unity and WebXR to maximize universal accessibility across browsers.
- Implemented real-time multiplayer systems through Photon Networking integration, enabling simultaneous user interaction.
- Designed and programmed a customizable avatar editor, increasing user retention and sense of identity within the platform.
- Managed the development lifecycle under Scrum Agile methodology, ensuring the technical delivery of critical milestones during the professional residency.
- Founded and led a software development initiative, managing the trademark registration process for “KreaVerse” before the IMPI (Instituto Mexicano de la Propiedad Industrial) and establishing the organization's operational and commercial foundations

High Tech Robotics School **Mérida Yucatán, México**
Video Game Development Instructor *August 2024 – January 2025*

- Provided technical training in C# programming and video game design in Unity for basic and intermediate levels.
- Directed the development of two complete video games as final course projects, supervising from the concept phase to final deployment.
- Trained students in the implementation of advanced game mechanics and optimization of user interfaces and experiences (UI/UX).

Marco Virtual MX **Remote (Based in Mérida, Yucatán, México)**

Metaverse Developer *May 2023 – August 2024*

- Programmed interactive mechanics and game logic for corporate metaverse environments using Unity and C#.
- Collaborated on the integration of XR technologies to create immersive experiences optimized for smooth execution on the web.
- Ensured technical performance of 60 FPS through script optimization and efficient 3D asset management on mobile devices.

EDUCATION

Universidad Tecnológica Metropolitana Mérida Yucatán, México

Bachelor of Science in Virtual Environments and Digital Business – Graduated 2026

Honors: Awarded for the highest GPA of the graduating class.

Universidad Tecnológica Metropolitana Mérida Yucatán, México

Associate Degree in Virtual Environments and Digital Business - Grduated 2024

Honors: Awarded for the highest GPA of the graduating class.

PROJECTS AND ACHIEVEMENTS

- **Balam Runner:** Lead developer for a 3D runner inspired by local culture, which earned third place in the GameJam Yucatán i6 2023.
- **NoTrash Adventure: EcoDash:** Lead developer for an educational video game focused on environmental ecology.
- **VARTICA Metaverse:** Web metaverse platform for artistic culture featuring customization and networking systems.
- **VR Logistics Simulator:** Developer of an immersive training experience for logistics with physical interaction mechanics, warehouse operations, and fire emergency simulation.

ADDITIONAL SKILLS

- Development of video games, metaverses, and immersive technologies (PC, mobile, and web); as well as multiplayer and XR systems.
- Testing, debugging, performance optimization, and API integration.
- Autonomous behavior programming (simple AI / state machines for NPCs and machinery) and procedural events.
- Mechanics iteration based on user testing and development of functional prototypes from concept.
- Interactive UI/UX implementation.
- Use of AI for development assistance.

TECHNOLOGIES

- **Game Dev & XR:** Unity (C#), WebXR, Photon Networking, ARFoundation, ARCore, Vuforia Engine, SteamVR, OpenXR (Meta Quest), Google Cardboard, Spatial Creator Toolkit, Godot Engine (GDScript).
- **Tools:** Git, Sourcetree, GitHub, Apache Superset, Jira, Asana, Notion, Microsoft Office.
- **Graphic Design and video editing:** Photoshop, Illustrator, Inkscape, Corel Draw, Filmora, Camtasia, Premiere, After Effects.
- **AI:** ChatGPT, Gemini, Grok, Claude.